ARNAB MANDAL

+91-9560452773 | arnabmandal2912@gmail.com | linkedin.com/arnabmandal2912 | github.com/arnabwithab | Portfolio

INTERNSHIPS & EXPERIENCE

Intern, Enterprise AI Platform, Fidelity International, Gurgaon Jun 2026 - Jul 2026

- Cut **PII** redaction latency to <**300ms** via **AWQ**-quantized fine-tuning of Gemma 4 on **AWS**, reaching **96% F1**.
- Shipped document sensitivity guardrail layer for enterprise AI gateway on **15k**+ monthly uploads, audited via **ELK**.
- Contributed in building internal AI Harness resulting in **22%** fewer tokens and **27%** faster generation.
- Built a DeepEval **harness** for RAG evaluation via the RAG triad; caught **3** groundedness regressions before release.

Freelance AI Engineer, value-based care healthcare client Feb 2026 - Apr 2026

- Deployed a multi-agent Text2SQL engine with sandboxed-agent execution serving **21** orgs and **200**+ users on AWS.
- Achieved **p50**<**3s** via tiered agent routing, auto-scaling Fargate, and semantic caching at **50**+ concurrent users.
- Engineered MySQL→S3→dbt→Postgres pipeline, with PITR: RPO<**5min**, RTO<**1h**, decoupled from live traffic.
- Built HNSW-indexed pgvector GraphRAG for schema discovery, achieving **0.87** context recall on domain knowledge.

AI Engineer Intern, DeepLure Research Pvt Ltd, Amravati Dec 2025 - Jan 2026

- Optimized inference service for **50**+ clients reducing per-request latency from 30ms to **2ms**.
- Migrated to TensorRT + Triton, achieving **8x** throughput, reducing AWS GPU costs by **$2,300**+/month.

Founding AI Intern, Critical AI Pvt Ltd, Bengaluru Jul 2025 - Sep 2025

- Deployed an on-prem RAG agent for a legal firm, cutting inference cost **30%** vs. cloud for **early adoption** of AI.
- Built a voice-primary multimodal RAG agent for geospatial queries, pitched to Indian Navy departments.

AI Intern, WESEE, Ministry of Defence, New Delhi May 2025 - Jul 2025

- Cut SME report prep time to **500ms** via deploying **3**+ MCP servers for agentic SQL analytics of radar/satellite data.
- Built LLM serving for internal MoD chatbot, used by **7**+ teams and handling **200**+ daily queries.

PROJECTS

Open-Source Contributor at vLLM: Domino Speculative Decoding

- Implemented Inference and Distributed training support for Domino, reporting a **18%** throughput gain on Qwen3.
- Improved acceptance length by **27%** over DFlash across 7 benchmarks, including **GSM8K** and HumanEval.
- Technology Stack: Python, PyTorch, Transformers, FSDP2

AutoLinks: Distributed Semantic Internal Linking Engine for SEO

- Architected async Goroutine workers with ElastiCache Streams queue for sitemap ingestion @ **150** articles/min over ECS.
- Utilized LLM-as-a-judge, achieving **85**+% acceptance, with **p95** Latency of **600ms**, lowering Gini coefficient by **20%**.
- Technology Stack: React, Go, chi, trafilatura, AWS, Qdrant, Clerk

Open-Source Contributions: Merged fixes in NVIDIA-NeMo, llm-d, OpenMetadata, WebMCP.

EDUCATION

Shiv Nadar University Noida, Delhi-NCR, India
B.Tech in Computer Science Engineering **(8.59/10.0)** *Aug 2023 - Jul 2027*

TECHNICAL SKILLS

Languages	Python, Go, TypeScript, CUDA, SQL
AI & ML	LangChain, LangGraph, LangSmith, LiteLLM, Optuna, MLflow
MLOps & Infra	vLLM, Triton, TensorRT, Ray Serve, Kubernetes, Docker, Jenkins, Prometheus
Cloud	GCP, AWS, Azure, Terraform

ACHIEVEMENTS

- **Smart SNU Hackathon'25 Winner**: Led a 6-member team to build and deploy the Krishi AI Sahayak app.
- **LeetCode Knight**: Achieved a **1,925 peak contest rating** and **solved 500+ DSA problems.**
- **Semifinalist speaker at SMVIT Parliamentary Debate'25**